

8(a)(i)	Read-off from Fig. 8.1, Speed of P-wave = 8.4 km s^{-1} Speed of S-wave = 4.8 km s^{-1} Distance travelled by P-wave = Distance travelled by S-wave $(t - 170) \times 8.4 = t \times 4.8$ $\Rightarrow t = 396.76 \approx 397 \text{ s}$	[1] for both correct speeds [1] for correct workings [1] for correct answer
8(a)(ii)	distance = $396.76 \times 4.8 = 1904.4 \approx 1900 \text{ km}$	[1]
8(a)(iii)	S-wave has <u>no velocity</u> in outer core; It cannot travel through outer core as <u>outer core is liquid</u>	[1] [1]
8(b)(i)	amplitude = 0.014 m frequency = $\frac{12}{2\pi} = 1.9098 \approx 1.9 \text{ Hz}$	[1] [1]
8(b)(ii)1	$v = \frac{ds}{dt} = -(0.014 \times 12) \sin(12t) \approx -0.17 \sin(12t)$	[1]
8(b)(ii)2	$a = \frac{dv}{dt} = -(0.14 \times 12^2) \cos(12t) \approx -2.0 \cos(12t)$	[1]
8(b)(iii)1	SHM $\Rightarrow a_o = -\omega^2 x_o $; $v_o = \omega x_o$; $\omega = 2\pi f$ $\therefore a_o = \omega(\omega x_o) = \omega v_o = 2\pi f v_o$ $PGA = 2\pi f v_o$	[1] for correct expression of a_o , v_o and ω [1] for correct working
8(b)(iii)2	$PGA = 0.90 \text{ m s}^{-2}$; $v_o = 0.081 \text{ m s}^{-1}$ $f = \frac{PGA}{2\pi v_o} = \frac{0.90}{2\pi(0.081)} = 1.7684 \approx 1.8 \text{ Hz}$ Using $v_o = \omega x_o$, $\Rightarrow 0.081 = 2\pi(1.7684)x_o$ $\Rightarrow x_o = 0.0072901 \approx 0.0073 \text{ m}$	[1] [1] [1]
8(b)(iii)3	Resonance takes place when the <u>external driving frequency of the wave from earthquake</u> is the same as its <u>natural frequency of the buildings</u> , Hence there is <u>maximum transfer of energy</u> , causng large amplitude	[1] [1]

8(b)(v)	Suitable test explained: $PGA \propto \frac{1}{r} \Rightarrow PGA = k \frac{1}{r}, \text{ where } k \text{ is a constant.}$ $k_1 = 56 \times 1.2 \approx 67$ $k_2 = 120 \times 0.53 \approx 64$ $k_3 = 220 \times 0.3 = 66$ Since k_1, k_2 and k_3 are similar, the equation is valid.	[1] [1] for three values of k. [1] for conclusion
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